

A REPORT
ON
**Generative Adversarial Network
Attack and Defense**

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GENERATIVE AI



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Chapter 1

Introduction

1.1 Abstract

Generative Adversarial Networks (GANs) can produce high-fidelity images, but their outputs can change when the latent input is perturbed. We compare basic and CMA-ES black-box latent attacks on DCGAN and Progressive GAN (PGAN). We also examine latent denoising and a reference-guided CMA-ES recovery method. LPIPS measures perceptual differences between paired images, while Fréchet Inception Distance (FID) compares image distributions. The reported results show stronger perceptual shifts under CMA-ES and lower reconstruction error after defense. The CMA-ES recovery step uses the clean original image as a reference, which limits its use when that image is unavailable.

1.2 Introduction

Over the last decade, deep generative models have achieved remarkable progress, with Generative Adversarial Networks (GANs) [1] emerging as one of the most successful paradigms for generating realistic images, audio, and video content. GANs consist of two neural networks, a generator and a discriminator, trained in a minimax game where the generator aims to produce samples indistinguishable from real data, and the discriminator attempts to distinguish between real and generated samples. This adversarial setup has led to the creation of models capable of producing high-resolution, detailed images that often fool human observers [5, 4].

GAN outputs can change under deliberate modifications to their latent inputs. Adversarial examples have been studied extensively for classifiers [9]; here we examine generator queries without access to model parameters or gradients. For example, an attacker might perturb a latent code sent to a public image-generation API.

These attacks have significant implications: an attacker could compromise the quality or semantics of generated content in sensitive contexts (e.g., synthetic medical images, ID document generation), or craft images that evade detection in GAN fingerprinting or forensic tasks.

This report investigates black-box attack strategies on two well-known GAN models: **DCGAN** [8], a pioneering convolutional GAN model, and **PGAN** [5], known for its progressive training and high-quality outputs. We implement and compare two attack strategies:

- A basic latent-space perturbation attack using random or structured modifications.
- An optimization-based strategy using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [2], a powerful black-box optimizer known for its sample efficiency and effectiveness in high-dimensional spaces.

To measure attack effectiveness, we rely on two primary metrics:

- **LPIPS** (Learned Perceptual Image Patch Similarity) [11] – a perceptual similarity metric that correlates well with human judgment.
- **FID** (Fréchet Inception Distance) [3] – a widely used metric to quantify the distributional distance between real and generated image sets.

Beyond attack strategies, we examine latent-space reconstruction using the original image as a reference. The recovery method does not retrain the original generator, but it requires that clean reference image during optimization [10].

Our experiments demonstrate that:

1. Both basic and CMA-based attacks can significantly alter GAN outputs in a black-box setting, with CMA-ES achieving up to 100% success.
2. Attacks on PGANs are slightly more potent, indicating possible architecture-level vulnerabilities.
3. The reported defense success rates range from 94% to 100% across the tested models and methods; the reference-guided optimization requires a clean original image.

Through extensive experimentation and analysis, this report provides a systematic evaluation of the adversarial robustness of GANs in black-box conditions, offering practical insights for both attackers and defenders in the deep generative modeling ecosystem.

Chapter 2

Black-box Adversarial Attacks

2.1 Black-Box Attack Methodology

In a black-box attack setting, the adversary has no access to the internal parameters, gradients, or architecture of the generative model. The attacker can only query the generator—typically by inputting latent vectors—and observe the corresponding outputs. This constraint reflects a realistic adversarial scenario in deployed systems, such as APIs serving pre-trained generative models.

In this work, we consider two black-box attack strategies: a simple, perceptual-difference-driven **basic perturbation attack**, and a more sophisticated **evolutionary optimization attack** using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES).

2.1.1 Basic Perturbation Attack

The basic attack leverages small random or structured changes in the latent vector fed into the generator. The core idea is that minor changes in the latent space may yield disproportionately large changes in the output image—especially in semantic features such as object orientation, facial attributes, or background composition.

Given an original latent vector $z \in \mathbb{R}^d$, the attacker generates a set of perturbed vectors $z' = z + \delta$, where δ is a small noise vector sampled from a uniform or Gaussian distribution. The corresponding image $G(z')$ is then compared with the original $G(z)$ using a perceptual similarity metric, such as LPIPS [11].

The attack is deemed successful if the perceptual distance exceeds a chosen threshold (e.g., $\text{LPIPS} > 0.1$), indicating that the visual appearance has significantly changed. This strategy does not rely on any gradient information, making it a true black-box approach.

Advantages:

- Simple and computationally inexpensive.
- Useful for identifying sensitivity regions in the latent space.

Limitations:

- Lacks directionality and efficiency—often requires many random samples.
- May fail to produce semantically meaningful or targeted perturbations.

2.1.2 Optimization-Based Attack: CMA-ES

To overcome the inefficiency of random perturbations, we adopt an evolutionary black-box optimization strategy using the **Covariance Matrix Adaptation Evolution Strategy (CMA-ES)** [2]. CMA-ES is a derivative-free algorithm designed to minimize (or maximize) an objective function by iteratively updating a population of candidate solutions based on their fitness values.

In our context, the attacker seeks a perturbed latent vector z^* that maximizes the perceptual distance $D(G(z^*), G(z))$, where D is the LPIPS score. The optimization objective is thus:

$$\max_{z^*} D(G(z^*), G(z))$$

The current attack implementation does not enforce an explicit bound on the distance between z^* and z .

CMA-ES begins with an initial population centered around the original latent vector z , and evolves it over multiple generations. At each iteration, new candidate vectors are sampled from a multivariate Gaussian distribution, and their fitness is evaluated via black-box queries to the generator. The covariance matrix of the sampling distribution is updated adaptively to steer the search toward regions of higher perceptual difference.

Advantages:

- Highly sample-efficient and effective in high-dimensional search spaces.
- Naturally suited for black-box settings with no gradient access.
- Produces large, perceptually meaningful changes in the image space.

Limitations:

- Computationally more intensive than random perturbations.
- Requires careful tuning of population size, step-size, and termination conditions.

2.1.3 Attack Success Criteria

We evaluate the attack success based on the LPIPS metric. Specifically, an attack is considered successful if the LPIPS score between the original and adversarial image exceeds a threshold (typically 0.1), indicating a noticeable perceptual shift. We also report the Fréchet Inception Distance (FID) between the set of original and adversarial images, to quantify the degradation in distributional quality.

This dual-metric evaluation allows us to capture both local perceptual shifts and global statistical divergence between generated image sets.

2.2 Attack Results and Analysis

To evaluate the effectiveness of the black-box attacks described previously, we conducted experiments on random latent vectors using two generator architectures: DCGAN and PGAN. For each sample, we applied both the basic perturbation attack and the CMA-ES optimization strategy. The success of an attack was measured using LPIPS (perceptual similarity) and FID (distributional quality).

2.2.1 Visual Comparison: Original vs Attacked Images

Figures 2.1 and 2.2 show side-by-side visual comparisons of original versus adversarial outputs for both DCGAN and PGAN, under Basic and CMA-ES attacks respectively. Each figure illustrates how significantly small latent perturbations can alter image content.

2.2.2 Attack Progression Analysis

To observe how attacks evolve over time, we tracked the LPIPS distance between original and adversarial outputs across iterations. Figures 2.3 and 2.4 show attack progression graphs for both DCGAN and PGAN.



FIGURE 2.1: Original vs Attacked images using Basic Perturbation Attack.

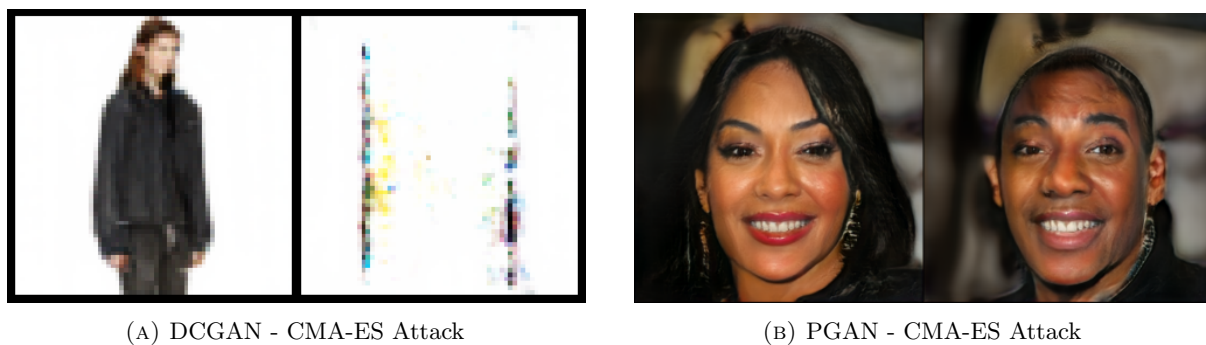


FIGURE 2.2: Original vs Attacked images using CMA-ES Attack.

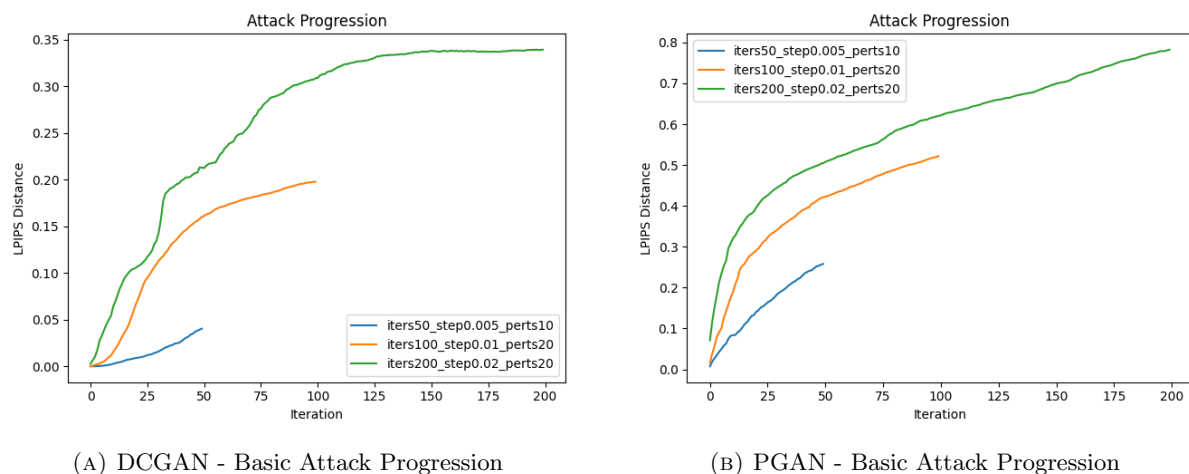
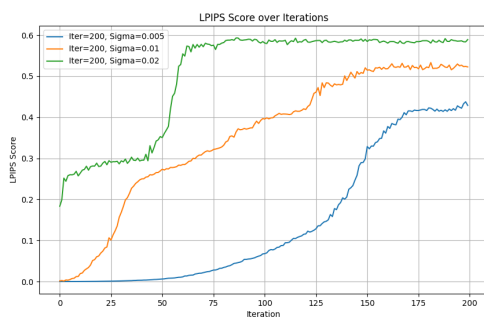


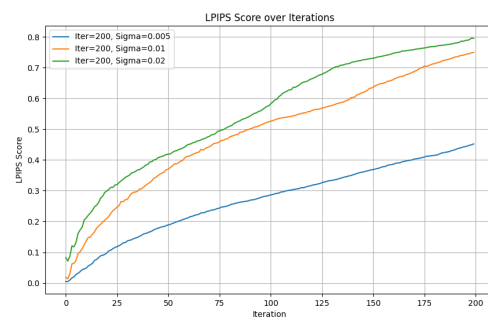
FIGURE 2.3: LPIPS over iterations for Basic Perturbation Attack.

2.2.3 Quantitative Summary of Attack Effectiveness

Table 2.1 presents a consolidated comparison of all quantitative metrics: Attack Success Rate, Average LPIPS (perceptual difference), and FID (distributional difference) for both the Basic and CMA-ES attacks across DCGAN and PGAN.



(A) DCGAN - CMA-ES Attack Progression



(B) PGAN - CMA-ES Attack Progression

FIGURE 2.4: LPIPS over iterations for CMA-ES Attack.

TABLE 2.1: Attack Metrics Across Models and Methods

Method	GAN Model	Success Rate (%)	Avg LPIPS	FID
Basic Attack	DCGAN	91.00	0.1881	70.49
	PGAN	97.00	0.1976	50.67
CMA-ES Attack	DCGAN	100.00	0.4228	119.76
	PGAN	100.00	0.4868	51.87

2.2.4 Interpretation of Results

The comparative analysis reveals several key insights:

- **CMA-ES has higher reported attack success** than the basic attack, reaching 100% on both tested generators.
- **Average LPIPS under CMA-ES is more than double** the corresponding basic-attack value for both tested generators, indicating a larger measured perceptual change.
- **FID changes differ by model.** The reported DCGAN value rises from 70.49 to 119.76, while the PGAN value rises from 50.67 to 51.87. FID alone does not establish which model is more robust.
- **PGAN has a smaller reported FID change** between methods, although its average LPIPS is higher under CMA-ES than under the basic attack.
- **The basic attack also meets the stated LPIPS threshold** for 91% of DCGAN samples and 97% of PGAN samples in the reported evaluation.

These comparisons describe the reported experiment; they do not establish behavior under other query budgets or perturbation limits.

Chapter 3

Defense

3.1 Defense Methodology

Black-box attacks on GANs perturb the latent vector input, resulting in semantically altered or degraded outputs. A robust defense must either (a) prevent these perturbations from affecting output or (b) recover the original image or latent code despite the perturbation. We explored several defense mechanisms with varying degrees of success.

3.1.1 Naive Defenses and Their Limitations

3.1.1.1 Gaussian Smoothing

A first-line approach involved applying Gaussian filters to the adversarial image $\hat{x}_{\text{adv}} = G(z')$ to remove high-frequency noise. However, attacks like CMA-ES do not operate in pixel space, and their perturbations manifest as semantic distortions rather than localized pixel-level noise. Thus, smoothing failed to restore original content and often blurred useful features, offering no perceptual or statistical improvement.

3.1.1.2 Defensive Distillation

Inspired by [7], we attempted defensive distillation on the GAN generator. A distilled generator G_{soft} was trained on outputs from the original G , intending to “compress” knowledge while learning to ignore adversarial examples. However, this approach was ineffective because:

- GANs do not produce discrete label outputs but high-dimensional image tensors.

- Distillation does not teach latent robustness unless specifically trained on adversarial variants, which defeats the black-box setting assumption.

Overall, pixel-space defenses failed due to the inherent design of latent-space attacks.

3.1.2 Latent Recovery: Reverse CMA-ES Optimization

The implemented reverse CMA-ES defense starts from the attacked latent vector z' and searches for an image close to a clean reference image $x_{\text{ref}} = G(z)$. This is a reference-guided recovery experiment: the clean image is supplied to the optimizer.

3.1.2.1 Formulation

Let $G : \mathbb{R}^d \rightarrow \mathbb{R}^{H \times W \times C}$ be the GAN generator. The implemented objective combines LPIPS and mean squared error (MSE) against the clean reference:

$$z^* = \arg \min_{\tilde{z}} \lambda \text{LPIPS}(G(\tilde{z}), x_{\text{ref}}) + (1 - \lambda) \text{MSE}(G(\tilde{z}), x_{\text{ref}}). \quad (3.1)$$

In the experiments, $\lambda = 0.5$. LPIPS is a learned perceptual metric [11]. CMA-ES evaluates candidate latents through generator queries without using generator gradients. The reference image makes the method unsuitable for a setting in which only the attacked latent or image is available.

Chapter 4

Auxiliary Latent Denoising Defense

Building upon the latent recovery defense via reverse CMA-ES, we further draw inspiration from **ZK-GanDef** [6], a recent defense framework that leverages GANs in a zero-knowledge adversarial training setting. ZK-GanDef introduces a method that does not require access to adversarial examples during training. Instead, it employs randomly perturbed inputs to train a discriminator that can distinguish between clean and adversarial data, thus improving robustness to a wide range of attacks.

4.1 Defense Strategy

4.1.1 ZK-GanDef Overview

ZK-GanDef proposes using a GAN architecture where the discriminator is trained to classify whether a sample is from clean or perturbed data, effectively acting as a defense-aware component. The generator attempts to produce examples that confuse the discriminator. Over time, this leads to a robust model without explicit use of adversarial training examples, reducing computational overhead and improving generalization.

4.1.2 Reverse CMA-ES as Latent Recovery

Our experiment uses latent recovery rather than the ZK-GanDef training method. Starting from the attacked latent z' , the reverse CMA-ES step searches for a latent vector whose generated image matches the clean reference x_{ref} :

$$z^* = \arg \min_{\tilde{z}} \mathcal{L}_{\text{rec}}(G(\tilde{z}), x_{\text{ref}}) \quad (4.1)$$

where $\mathcal{L}_{\text{rec}}$ is a perceptual similarity metric (e.g., LPIPS [11]).

The optimizer requires the clean reference image. The method therefore should not be interpreted as zero-knowledge recovery.

4.1.3 Auxiliary Generative Network Initialization

Motivated by ZK-GanDef’s use of a learned component (the discriminator), we introduce an *Auxiliary Generative Network* (AGN) f_θ which attempts to estimate the original latent vector z from the adversarial latents:

$$\hat{z}_0 = f_\theta(\hat{z}_{\text{adv}}) \quad (4.2)$$

The auxiliary model is trained on synthetic noisy/clean latent pairs. For a sampled latent z and perturbation δ , its encoder predicts the clean latent from $z + \delta$. Its training includes the reconstruction term

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{z,\delta} [\|f_\theta(z + \delta) - z\|_2^2] \quad (4.3)$$

The implementation also uses an adversarial discriminator loss. The encoder output initializes the reference-guided CMA-ES search. The intended benefits are:

- **Faster convergence:** AGN reduces the search space CMA-ES needs to explore.
- **Higher fidelity:** The final recovered image $G(z^*)$ has lower perceptual distance to the clean target, improving defense metrics.

4.1.4 Manifold Perspective and Limitations

For small perturbations, a denoising model may provide a useful initialization near the clean latent. The reverse CMA-ES stage then searches for a generated image close to the supplied reference. The reported comparisons evaluate this combination on DCGAN and PGAN; they do not establish recovery when the reference image is unavailable.

The results should be interpreted as a comparison of the tested methods under the stated reference-guided setup.

Connection to ZK-GanDef The auxiliary model uses synthetic latent perturbations rather than CMA-ES attack examples during training. That is a conceptual connection to ZK-GanDef, but the reference-guided optimization stage is a different method.

Improvement to Reverse CMA-ES The auxiliary initialization is intended to help CMA-ES in two ways:

- **Initialization:** The search starts from an estimated clean latent rather than directly from the attacked latent.
- **Reconstruction:** The reported comparison tests whether this initialization reduces LPIPS to the clean reference.

Together, these stages combine denoising on synthetic noisy/clean latent pairs with reference-guided reverse CMA-ES. They do not demonstrate a defense that operates without a clean reference at test time.

4.2 Defense Results and Analysis

To evaluate the effectiveness of the defense strategies described above, we conducted experiments on random latent vectors using two generator architectures: DCGAN and PGAN. For each sample, we first applied the CMA-ES optimization attack, and then used both our defense strategies. The success of an attack was measured using LPIPS (perceptual similarity).

4.2.1 Visual Comparison

Figures 4.1 and 4.2 show the original, attacked, and defended outputs for DCGAN and PGAN. These examples illustrate the reported results; success rates are summarized separately in Table 4.1.

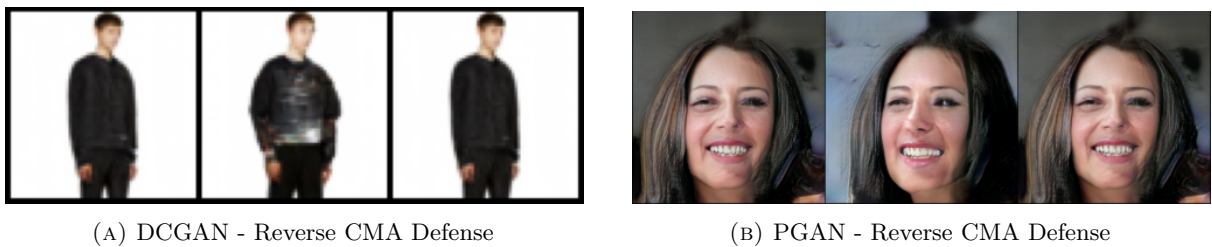


FIGURE 4.1: Defense using Reverse CMA Defense.

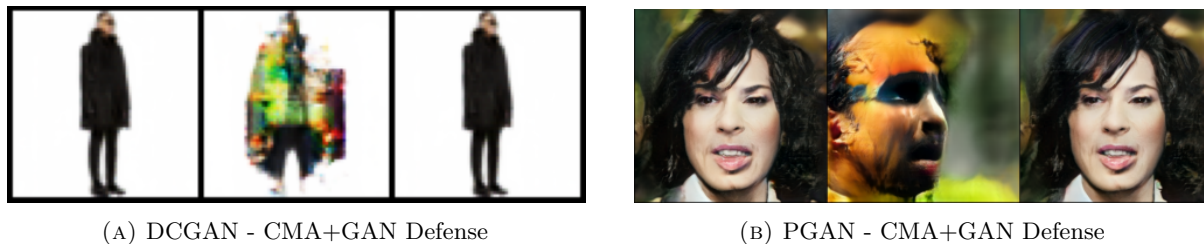


FIGURE 4.2: Defense using an auxiliary GAN alongside reverse CMA-ES.

4.2.2 Defense Progression Analysis

To observe how defense evolves over time, we tracked the LPIPS distance between original and defense outputs across iterations. Figures 4.3 and 4.4 show defense progression graphs for both DCGAN and PGAN.

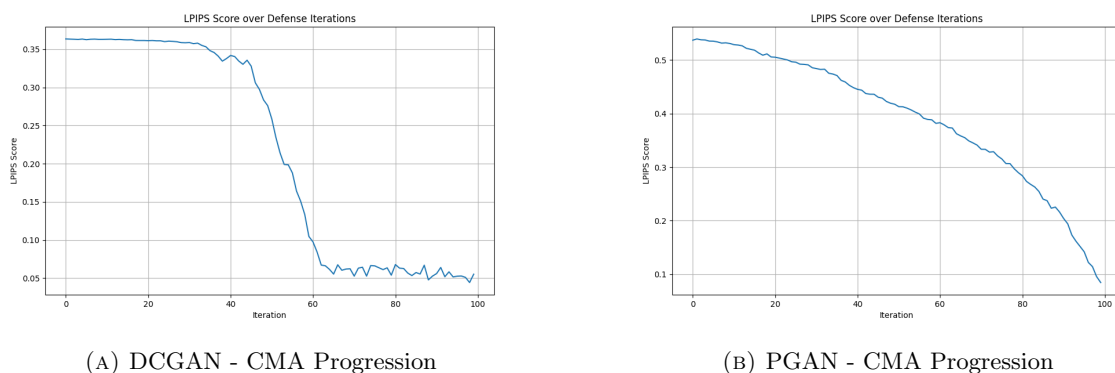


FIGURE 4.3: LPIPS over iterations for Reverse CMA Defense.

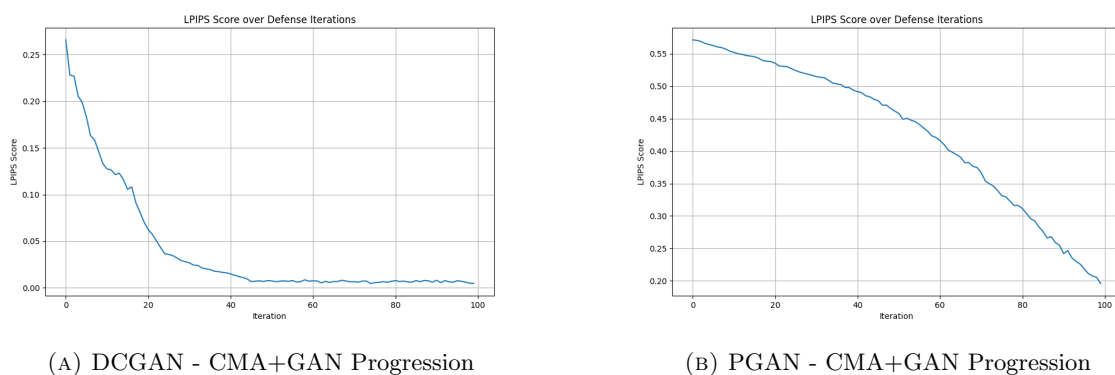


FIGURE 4.4: LPIPS over iterations for an auxiliary GAN alongside reverse CMA-ES.

4.2.3 Defense Evaluation and Results

TABLE 4.1: Reported defense results after CMA-ES attacks on DCGAN and PGAN

Defense	Model	Success (%)	LPIPS to reference	LPIPS improvement
CMA baseline	DCGAN	95.00	0.0204 ± 0.0158	0.3924
CMA baseline	PGAN	94.00	0.0659 ± 0.0233	0.4091
CMA + GAN	DCGAN	100.00	0.0148 ± 0.0097	0.3950
CMA + GAN	PGAN	99.00	0.0300 ± 0.0164	0.4504

Defense	Model	Recovered FID	FID improvement
CMA baseline	DCGAN	22.9478	96.3531
CMA baseline	PGAN	6.6100	51.4246
CMA + GAN	DCGAN	18.4019	97.5782
CMA + GAN	PGAN	1.8398	50.0304

The source table gave both 95% and 100/100 for the DCGAN CMA baseline. Its percentage is retained here; the conflicting count is omitted pending verification against sample-level results.

4.2.3.1 Insights and Analysis

From Table 4.1, several conclusions emerge:

- **CMA+GAN consistently outperforms the CMA baseline** across both generators (DCGAN and PGAN), showing improved defense success rates and perceptual fidelity.
- **Defense success rate:** The reported CMA baseline rates are 94% or higher. The reported CMA+GAN rates are 100% on DCGAN and 99% on PGAN.
- **LPIPS:** CMA+GAN has lower reported LPIPS to the clean reference than the CMA baseline for both models.
- **FID:** The recovered-image FID values are lower for CMA+GAN in the reported comparisons.
- **Scope:** Both tested generators show improvements under this reference-guided evaluation. These results do not establish performance without a clean reference.

LPIPS Threshold Visual Comparison

We visualize the LPIPS score distributions before and after applying the defense for both DCGAN and PGAN models. These comparisons use a threshold of LPIPS > 0.1 to evaluate the defense effectiveness. The x-axis represents image indices, and the y-axis represents LPIPS scores.

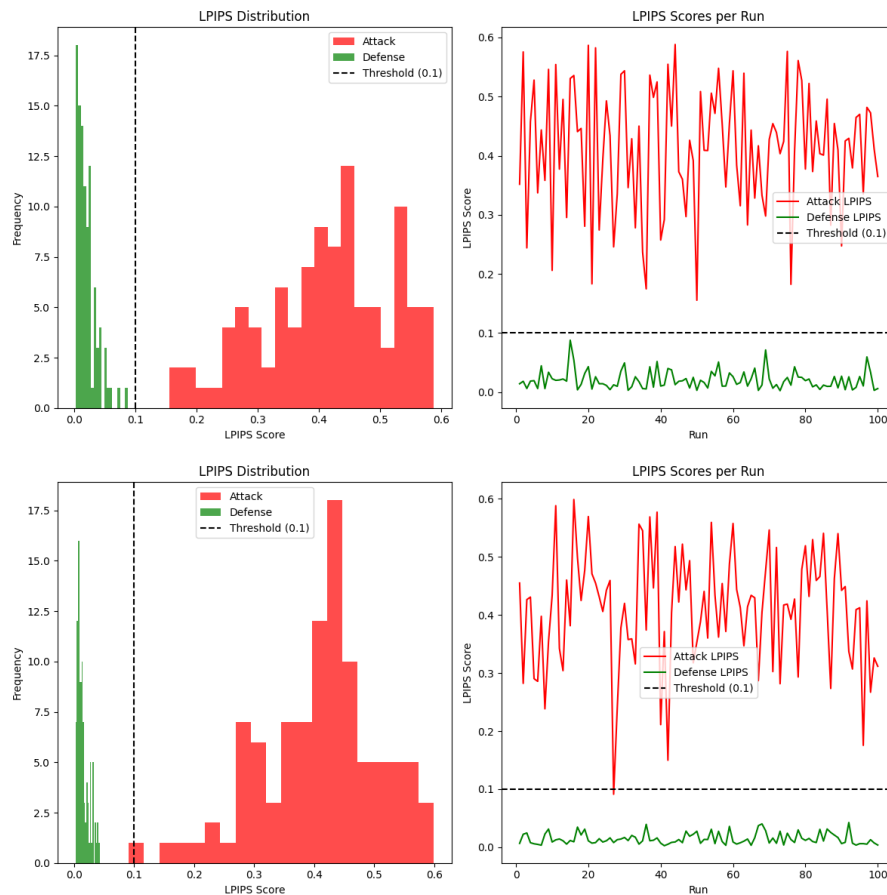


FIGURE 4.5: LPIPS Comparison for DCGAN. Top: CMA Baseline. Bottom: CMA+GAN Defense.

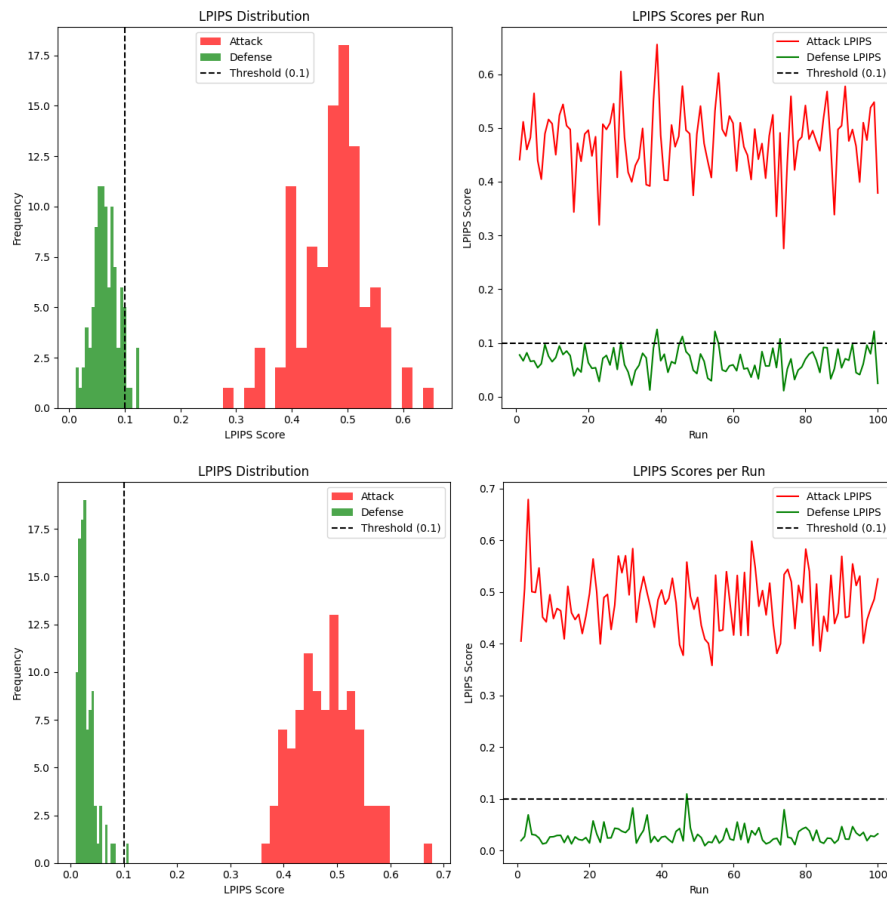


FIGURE 4.6: LPIPS Comparison for PGAN. Top: CMA Baseline. Bottom: CMA+GAN Defense.

Chapter 5

Conclusion

In this work, we investigated the efficacy of black-box adversarial attacks and their corresponding defenses within the latent space of generative models. We evaluated two main attack methodologies — a basic perturbation-based approach and a more sophisticated black-box optimization attack using Covariance Matrix Adaptation Evolution Strategy (CMA-ES). Our experiments demonstrated that CMA-ES, while computationally heavier, produces more effective and perceptually impactful adversarial samples, particularly when attacking both DCGAN and PGAN architectures.

We then explored several defense strategies, including smoothing and distillation. The reverse CMA-ES experiment optimizes a latent vector against the clean original image supplied as a reference. This supports an evaluation of reference-guided recovery, rather than recovery from the attacked input alone.

We also used an **Auxiliary Generative Network (AGN)** to initialize the CMA-ES optimization. In the reported comparisons, the combined method had lower LPIPS to the clean reference and lower recovered-image FID than the CMA baseline on both tested generators. The tabulated results need sample-level verification before stronger performance claims can be made.

The auxiliary model is conceptually related to **ZK-GanDef** because it trains on perturbed inputs, but this experiment uses a different, reference-guided test-time method. The clean-reference requirement is the main limitation for deployment without access to the original image.

Ultimately, this work highlights the importance of leveraging generative priors and structured latent spaces for both attack and defense in adversarial machine learning.

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